Making the World's Fastest Racket Sport even Better: A Systematic Review of Artificial Intelligence-based Objective Player Performance Assessment in Badminton

Christian Arthur
The University of New England
Raymond Chiong
rchiong@une.edu.au
The University of New England
Fareed Ud Din
The University of New England
Farshid Hajati
The University of New England
Subrata Chakraborty
The University of New England
David Paul
The University of New England
Mitchell Welch
The University of New England
Robert G. Crowther
The University of New England

Systematic Review

Keywords: Artificial intelligence, badminton, computer vision, player assessment, sports analytics

Posted Date: March 18th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9105158/v1

Additional Declarations: No competing interests reported.

Abstract

Artificial intelligence (AI) is increasingly promoted as a means of providing objective player performance assessment in badminton. Compared to other sports, however, the supporting evidence remains fragmented. A systematic review based on 51 studies that satisfied the established quality and eligibility criteria from three major databases (covering the period 2018 to the end of 2025) reveals four dominating methodological schools: computer vision stroke tracking, movement-pattern recognition, spatio-temporal analysis of rally sequences, and multi-modal frameworks that integrate several data streams. Although many studies report high classification or prediction accuracy, only a small proportion of them employ shared validation datasets or evaluate repeatability across testing sessions, which limits the generalisability of their findings. Common shortcomings include small or imbalanced data samples, weak alignment with established sport-science theory, substantial computational requirements, and participant pools drawn largely from elite athletes in a single geographic region. Recent work has begun integrating explainable AI with retrieval-augmented generation (RAG) and large language model (LLM) frameworks to provide grounded, query-responsive feedback that links visual detections and performance metrics to structured match evidence. Future research should focus on larger and more diverse datasets, alignment with skill development models, transparent output formats, and validation across competitive levels and contexts with these state-of-the-art explainable AI-based RAG or LLM frameworks.

1. Introduction

Since its inclusion in the Olympic Games in 1992, badminton has developed into a highly competitive racquet sport with a well-established professional structure [1]. The sport is widely recognised as the fastest racquet sport, characterised by extremely rapid rallies and shuttle speeds that demand quick reactions, precise technical execution, and continuous tactical adaptation. It is played recreationally by an estimated 200 million people worldwide, with more than 7,000 athletes competing annually in hundreds of national and international events [2].

Traditionally, coaches, selection committees, and technical administrators have based badminton player assessment on subjective evaluation [3, 4]. While capitalising on the domain expertise of the practitioner, this approach can lead to inconsistent and biased approaches to player development, competitive structuring, and participant satisfaction [5, 6]. The subjective nature of traditional assessment models has spurred increasing interest in more objective, data-driven approaches to player evaluation [7].

Compared to other sports, badminton is a complex one to assess because of its multiple technical requirements, fast pace of play, and the presence of multiple skill components [8, 9]. To be proficient in multiple domains, players must have demonstrated proficiency in stroke techniques, court movement, tactical decision-making, and competitive temperament [10, 11]. However, these skills are integrated at speeds much faster than other racket sports [12].

In most countries worldwide, club-level badminton associations use classification systems to divide players into different skill categories to help establish appropriate competitive structures and training organisation [13]. Despite this, the subjectivity of assessment often undermines the effectiveness of these systems, resulting in inconsistent assessment, player disputes, and the ability to manipulate for competitive advantage [14–16].

The integration of artificial intelligence (AI) technologies provides promising pathways to address these challenges by providing a systematic, data-driven multi-factor analysis of player performance [17]. With recent advances in computer vision, machine learning, and data analytics, opportunities exist for more objective assessment frameworks [18]. However, such systems are theoretically incomplete, methodologically unexplored, and empirically untested [19].

This review studies AI approaches for badminton player performance assessment. It evaluates their reported accuracy and practical utility, identifies recurring methodological limitations, and highlights underexplored aspects of performance data. By charting prevailing trends and research gaps, we seek to steer future investigations towards rigorous, generalisable, and practitioner-oriented AI frameworks for objective skill assessment in badminton. To the best of our knowledge, this is the first such review on AI-based badminton player performance assessment in the relevant literature.

2. Methods

This study follows a systematic literature review methodology guided by the PRISMA framework [20]. The review protocol adapts the procedures outlined by Helbach et al. [21], beginning with the definition of the study’s scope and objectives. Specifically, the review addresses three research questions (RQs):

RQ1. Which AI approaches are currently applied to badminton player assessment, and what evidence supports their effectiveness?

RQ2. What theoretical, methodological, and technical constraints are most frequently reported across these studies?

RQ3. How are AI assessment results communicated to ensure they are clear and trusted by coaches and athletes?

These questions structure the subsequent stages of database searching, study screening, data extraction, and synthesis.

2.1. Search Strategy

The keywords , presented in Table I, were selected to reflect three core themes: the sport context (e.g., “badminton”, “racket sport”), type of analysis (e.g., “pose estimation”, “performance assessment”), and computational approaches (e.g., “machine learning”, “computer vision”). This structure helped capture a comprehensive yet targeted set of publications relevant to the review’s objectives.

Table I Final search queries used across selected academic databases.

Database	Final query
Web of Science Core Collection	*TS=((badminton OR "racket sport*" OR "racquet sport*") AND (video analysis OR motion tracking OR pose estimation OR action recognition OR movement analysis OR "performance assessment") AND (artificial intelligence OR machine learning OR deep learning OR neural network* OR computer vision))*
Scopus	*(badminton OR "racket sport*" OR "racquet sport*") AND ("video analysis" OR "motion tracking" OR "pose estimation" OR "action recognition" OR "movement analysis" OR "performance assessment") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network*" OR "computer vision")*
IEEE Xplore	*(("badminton" OR "racket sports" OR "racquet sports") AND (("artificial intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "neural networks")) AND (("player assessment" OR "skill classification" OR "performance metrics" OR "technical analysis" OR "sports analytics")))*

2.2. Eligibility Criteria and Study Selection

The initial screening was done by the first and fourth authors. Once the screening was completed, all articles that were not related to the topic were excluded from further consideration. For articles lacking a clear title and abstract, full-text screenings evaluated the content to determine whether they should be included. Decisions about which articles to include were made after careful full-text evaluations, with all authors in full agreement. Figure 1 illustrates the data extraction process, providing a clear overview of how the 356 records retrieved from the academic databases were processed to result in the 51 studies that are included in the data synthesis and analysis of this review.

Eligibility criteria were established to ensure the relevance of the review to AI-based badminton player assessment. Included studies were peer-reviewed journal articles or conference proceedings published between January 2018 and December 2025 (note: a few early-access articles dated 2026 were also inlcuded) in English. Exclusions applied to abstract-only entries, non-English publications, and studies before 2018. Articles had to focus on AI applications for player assessment, skill evaluation, or performance classification, with methodological clarity and empirical implementation. Studies on tactical analysis, match prediction, or non-AI approaches were excluded unless they addressed player-level assessment. Relevant literature was identified through systematic searches in IEEE Xplore, Scopus, and Web of Science.

3. Results

3.1. Overview of AI Methods Applied to Badminton Player Assessment (RQ 1)

3.1.1. Study Characteristics

Figure 2 illustrates the yearly publication trend on AI applications for badminton player assessment. Research output was low from 2018 to 2020, with one publication per year. A clear increase can be seen in 2022, rising to ten publications and remaining similar in 2023 and 2024. The number of studies then increased sharply to nineteen in 2025, indicating accelerated research activities in the field. This trajectory indicates growing scholarly interest coinciding with the maturation of deep-learning toolkits that lower technical barriers to entry.

3.1.2. Methodological Approaches

Table II. Methodological approaches employed in AI-based badminton player assessment studies.

No.	Study	Study Focus	Player Sample	Data Source(s)	AI Methodology Applied	Performance Metrics Reported	Assessment Framework Orientation	Validation Approach
1	[22]	Stroke-level dataset for tactical performance prediction	Elite athletes	Video Footage	Transformers, Bidirectional-Gated Recurrent Unit (GRU), Graph Models	Area Under the Curve (AUC), Accuracy	Tactical; Post-hoc	Task Benchmarks
2	[5]	Automatic point and stroke annotation	Elite athletes	Video Footage	Convolutional neural network (CNN) and Support Vector Machine (SVM)	Accuracy, Mean Average Precision@0.5, Edit Score	Technical and Tactical; Post-hoc	Split-based Evaluation
3	[6]	Profiling performance via score progression	Elite athletes	Match Data	SVM, RF, K-means	R², Accuracy, Std. Dev	Technical; Post-hoc	Train/Test Spl
4	[7]	Prediction and tactical decision analysis in women's singles	Elite athletes	Video Footage	DT, RF, XGBoost, SVM	Accuracy (up to 87.5%)	Technical and Tactical; Post-hoc	Cross-validation (90/10 split)
5	[8]	Closed-loop AI for imitation, simulation, and strategy optimisation	Elite athletes	Video Footage (ShuttleSet, annotated datasets)	RallyNet, Simulation Environment	Loss metrices, Win Rate	Tactical; Real-time and Post-hoc	Benchmarking vs baselines (ShuttleNet and DyMF) in simulated matches
6	[9]	Behaviour analysis from broadcast videos with visual analytics	Elite athletes	Video Footage	You Only Look Once (YOLO)v3 Model, OpenPose, ResNet-18, LSTM	Accuracy (97%), Precision, Recall, F1 (hit-frame detection)	Tactical; Real-time and Post-hoc	Split-based Evaluation (train/val/test; semi-supervised learning)
7	[10]	Accuracy assessment of forehand smash biomechanics	Collegiate athletes	Video Footage (front-view rallies, Mediapipe extraction)	Skeletal Pose Estimation (MediaPipe), Dynamic Model, RF, Ridge	Accuracy (up to 97.4%), Precision, Recall, F1	Technical; Post-hoc	Train/Test Split (70/30), Cross-validation, Expert validation
8	[11]	Deep learning-based recognition of badminton actions	Recreational athletes	Hybrid Approach	Long Short-Term Memory (LSTM), Continuous Learning	Accuracy (63%, 84%, 92% across datasets)	Technical; Post-hoc	Benchmark on HKSR and NTU skeleton datasets
9	[12]	Detection and classification of six-corner footwork	Recreational athletes	Video Footage	YOLOv8, YOLOv9	Mean average precision (0.633 vs 0.605), Precision, Recall, F1	Technical; Real-time	Train/Val/Test Split (80/10/10); Model comparison
10	[13]	Shot and strategy recognition using wearable sensors	Elite and semi-professional athletes	Wearable Sensors	CNN, LSTM	Shot Accuracy (90.9%), Strategy Accuracy (80%), Precision, Recall, F1	Technical and Tactical; Post-hoc	Train/Val/Test Split (64/20/16), Class weighting
11	[17]	Action recognition using spatio-temporal skeleton features	Youth athletes	Video Footage (3D skeletal data via MediaPipe)	Weighted Ensemble Machine Learning Models	Accuracy (95.38%), Precision, Recall, F1	Technical; Post-hoc	5-fold Cross-validation

12	[18]	Detection and classification of six badminton strokes	Elite athletes	Video Footage (Kaggle badminton_stroke_video, 3000+ annotated images)	YOLO-HGNet	mAP (96.1%), Accuracy (95.4%), Precision, Recall, F1	Technical; Real-time and Post-hoc	Benchmark vs YOLOv5, OpenPose, CatBoost, XGBoost
13	[23]	Footwork recognition and 3D trajectory extraction	Sub-elite athletes	Video Footage (binocular cameras)	Faster R-CNN Binocular Positioning	Shoe localisation accuracy (97.2%) and Positioning error (0.129 m)	Technical; Real-time and Post-hoc	Train/Val/Test Split (90/10), Trajectory validation
14	[24]	Player evaluation using DRL with technical and tactical contexts	Elite athletes	Video Footage (BWF match videos, 2018–2020)	DRL (LSTM-based Q-function), AlphaPose, TrackNet	Action Value (Q-values), Correlation with Score/Rank	Technical and Tactical; Post-hoc	Baseline comparisons, Spearman Correlation
15	[25]	Shot prediction and classification from broadcast videos	Recreational athletes	Video Footage (26 broadcast videos, Shuttlecock Trajectory Dataset)	YOLOv5, DeepSORT, CNN	MOTA (73%), HOTA (72%), IDF1 (73%)	Technical; Post-hoc	Benchmark with trajectory metrics
16	[26]	Classification of badminton shots (lob, smash, net)	Elite athletes	Video Footage (BWF Men's Singles matches)	Keras-Mediapipe, YOLO-NAS (Neural Architecture Search)	Accuracy (73.5%), Precision, Recall, F1, AUC	Technical; Post-hoc	Train/Val/Test Split (70/10/20), Confusion Matrix
17	[27]	Predicting shot accuracy using Quiet Eye and biomechanics	Elite, Intermediate, and Novice athletes	Hybrid Approach (Eye-tracking + Motion Capture)	Neural Network, SHapley Additive exPlanations (SHAP)	Accuracy (85.7%), Precision (88.3%), Recall (83.1%), F1 (0.856), AUC (0.823)	Technical; Real-time and Post-hoc	Train/Val/Test Split (70/15/15), Ablation Study
18	[28]	Coaching framework for stroke, stance, and scoring	Professional, Intermediate, and Novice athletes	Wearable Sensors (IMU: wrist, palm, left leg, right leg)	CNN (stroke classification, score regression), kNN (stance error estimation)	Stroke Accuracy (89.1%), Performance R^2 (0.888), Success Rate (%)	Technical; Post-hoc	Cross-person validation, Train/Val/Test Split (60/20/20), Baseline comparisons
19	[29]	Shot influence prediction using long/short-term dependencies	Elite athletes	Video Footage (43,838 shots, 4,350 rallies, 75 matches, 2018–2021)	CNN (short-term), Transformer (long-term)	AUC (0.871), Brier Score (0.143), Efficiency gains	Tactical; Post-hoc	5-fold Cross-validation
20	[30]	Smash detection using optimised deep learning model	Elite athletes	Video Footage (YouTube broadcast matches)	Residual-Shuffle Net + Upgraded Pufferfish Optimiser (UPO)	Accuracy (96.4%), Precision (0.95), Recall (0.87), F1 (0.87)	Technical; Post-hoc	Train/Test Split (85/15), Baseline comparisons (AlexNet, GoogleNet, ResNet-18, I3D)
21	[31]	Pose estimation with improved YOLOv8-Pose and attention	Recreational athletes	Video Footage (xBHPE dataset, Kinect v2, 4000 samples)	YOLOv8-Pose + Local Attention	Mean squared error (12.72), Percentage of keypoints @0.2 (0.7793), Frame per second (66.4)	Technical; Real-time and Post-hoc	Benchmark vs HigherHRNet, BlazePose, LitePose
22	[32]	Biomechanics, injury prevention, and tactical analysis	Elite athletes	Hybrid Approach (Video, Wearable Sensors, Historical Match Data)	CNN (pose estimation)	CV Models (93% accuracy), Injury Prediction (90%), RL Defence Gain (20%)	Technical and Tactical; Real-time and Post-hoc	Cross-validation, Baseline comparisons, Empirical training tests
23	[33]	Automated detection of badminton players from	Elite athletes	Video Footage (YouTube broadcast matches: 2011 All England, 2012 Olympics, 2017 Asia Championship)	Faster R-CNN	Average Precision (PR curve-based)	Technical; Post-hoc	Case-based training/testing on single and combined videos

		broadcast videos						
24	[34]	Autonomous motion tracking of squash players	Elite athletes	Video Footage (broadcast squash matches, PSA Canary Wharf Classic 2013)	Multi-person Pose Estimation (CNN),	R² (0.99), Error in Distance (3.73%), Speed Error (1.26%)	Technical and Tactical; Post-hoc	Benchmark vs manual tracking, Filtering vs Unfiltered data
25	[35]	3D shuttle trajectory reconstruction from monocular video	Elite athletes	Video Footage (TrackNetV2 dataset + 40 YouTube matches)	Graph-based court detection, U-Net, GRU	Hit Detection (94.6% F1), 3D Reconstruction Error (~8 cm synthetic)	Technical and Tactical; Post-hoc	Benchmarks on TrackNetV2 Synthetic Trajectory Evaluation
26	[36]	Survey of video action recognition in sports (datasets, methods, applications)	Multiple sports (team & individual)	Video Footage (SoccerNet, Badminton Olympic, Diving48, FineGym, TTStroke-21, etc.)	2D/3D CNN, Two-stream, Transformer, Skeleton-based	-	Technical and Tactical; Post-hoc	Comparative analysis across datasets and models
27	[37]	Court line extraction from broadcast badminton videos	Elite athletes	Video Footage (BadmintonWorld.tv tournament videos)	Horizontal Line Projection + K-means, Morphological Corner Detection	Pixel Error (0–3 px), Frame Rate (33.3 FPS)	Technical; Real-time and Post-hoc	Comparative benchmark vs Hough Transform methods
28	[38]	Motion trajectory tracking with improved KCF and depth fusion	Elite athletes	Video Footage (binocular cameras, badminton arm strokes)	ViBe (background subtraction), KCF + Depth Fusion	Tracking Success Rate (98%), Trajectory Error (0.365), Speed Error (0.116)	Technical; Real-time and Post-hoc	Benchmark vs Kalman Filter & Original KCF
29	[39]	Stroke recognition in badminton using TSN with attention	Elite athletes	Video Footage (badminton stroke videos)	Temporal Segment Network (TSN) + Lightweight Attention	Recall (91.2%), Accuracy (91.6%)	Technical; Post-hoc	Train/Test Split
30	[40]	Shot refinement by combining shuttle tracking and hit detection	Elite athletes	Video Footage (BWF matches, 32 + 354 matches, 8,975 + 98,675 frames)	TrackNet (shuttle), YOLOv7 (swing), DensePose (pose), Shot Refinement Algorithm (SRA),	Shot Detection: Precision 0.897, Recall 0.913, F1 0.905; Shot Classification: Accuracy 72.1%	Technical and Tactical; Post-hoc	Benchmark vs TrackNet; Train/Val/Test split; t-IoU evaluation
31	[41]	Shuttlecock detection with lightweight small-object detector	Elite athletes	Video Footage (custom shuttlecock dataset, 6113 train / 1586 test images)	YOLO-inspired CNN	Mean Average Precision (98% shuttlecock), FPS (30 on Jetson Nano)	Technical; Real-time	Benchmark vs YOLOv3/v4/v5, v3-tiny, v4-tiny on public and shuttle datasets
32	[42]	Rally outcome prediction using stroke sequences	Elite athletes	Video Footage (London 2012 Olympics, 10 annotated players, 498 rallies)	ResNet-18, Bidirectional LSTM, Faster CNN	Accuracy (0.70), Brier Score (0.30)	Tactical; Post-hoc	Train/Test Split (398/100 rallies)
33	[43]	Stroke detection using wearable wrist device	Recreational athletes	Wearable Sensors (Arduino Nano 33 BLE, accelerometer, gyroscope)	CNN (embedded), Motion segmentation	Accuracy (100% in small dataset), Confusion Matrix	Technical; Real-time	Prototype testing on limited stroke dataset
34	[44]	Optimisation of doubles positioning and movement	Elite athletes	Match Data (2019 BWF Super Series finals, 10 matches)	Numerical Modeling (7 nonlinear equations, MATLAB fsolve)	Optimal attack & defense zones, Running circle model	Tactical; Post-hoc	Simulation and validation with BWF match statistics
35	[45]	Kinect based posture recognition and action evaluation system	Badminton athletes and collegiate participants	Multi Kinect V2 skeletal and depth data	Sensor calibration, Kalman filtering, CNN with attention, Transformer	Accuracy up to 98.75%; MAE 0.1364; F1 score 0.6911; 9.27 ms processing time	Technical; Real-time and Post-hoc	Baseline comparison and ablation study

36	[46]	Vision-based badminton stroke recognition and tactical analysis	Elite badminton athletes	Broadcast and match video footage	Improved YOLO-based detection, pose estimation, temporal sequence modelling	Stroke classification accuracy above 90%; mAP and F1-score reported	Technical and Tactical; Post-hoc	Train/Test split, baseline comparison with standard YOLO variants
37	[47]	Vision-based badminton posture recognition and training assistance system	Badminton players (experimental evaluation cohort)	Improved OpenPose (MobileNet backbone) + Particle Filter tracking	Improved OpenPose with MobileNet backbone; Particle Filter tracking	FPS, Recall, Precision, Accuracy, F1-score, response time, CPU usage	Technical; Real-time	Experimental evaluation and model comparison
38	[48]	Hybrid vision–language framework (ChatMatch) for badminton video understanding	Professional players and coach (evaluation); elite match videos	Broadcast match videos (20 singles matches)	UNet (court segmentation), YOLOv5, ResNet-50 (action recognition), GPT-3.5-based multi-agent system	Location accuracy 0.991; Action accuracy 0.902; Gesture accuracy 0.950	Technical and Tactical; Post-hoc	Train/Test split; confusion matrices; user-based evaluation.
39	[49]	Intelligent badminton handle with multinode MEMS sensors for explainable motion recognition	Badminton players (experimental evaluation cohort)	Multinode MEMS sensors embedded in racket handle (accelerometer and gyroscope data)	CNN-based motion recognition with explainability analysis	Recognition accuracy above 90%; precision, recall, F1-score reported	Technical; Real-time	Experimental validation with controlled stroke trials; comparative evaluation with baseline models
40	[50]	Inertial sensor–based badminton swing recognition	10 players; 1,200 swing samples	Racket-mounted accelerometer and gyroscope data; VideoBadminton dataset	PCA, SVM (grip classification), AdaBoost (six-class swing recognition)	Accuracy 95.93%; Precision, Recall, F1-score; AUC up to 0.97	Technical; Post-hoc	5- and 10-fold cross-validation; baseline comparison
41	[51]	Grip-force analysis for badminton performance evaluation using flexible pressure sensors	30 male participants (15 beginners, 15 national-level athletes)	Flexible piezoresistive pressure sensors embedded in racket handle	Signal acquisition and force-feature extraction (Imax, Iave, T); statistical correlation analysis	Grip peak force, mean force, duration; performance scores; correlation coefficients (r ≈ −0.6 to −0.7)	Technical; Post-hoc	Group comparison (beginner vs athlete); correlation analysis; significance testing ($p < 0.05$)
42	[52]	Deep learning and analytics for biomechanics, injury prediction, and tactical modelling	Competitive badminton players	Match video, wearable sensors, historical data	CNN pose estimation; Random Forest (injury).	93% biomechanical accuracy; 90% injury prediction; 15% speed gain; 25% injury reduction	Technical and Tactical; Real-time and Post-hoc	Controlled evaluation, confusion matrix, tactical performance comparison
43	[53]	Motion tracking and decomposition model for badminton training	Badminton training participants	3D limb joint coordinate data via motion sensing system	3D decomposition model; Gauss-Newton optimisation	Accuracy up to 99.31%; >34% improvement vs LSTM; 32.5% faster response	Technical; Real-time	Baseline comparison with LSTM; model accuracy and efficiency tests
44	[54]	Vision-based badminton stroke detection and performance evaluation	Competitive badminton players	Match video footage	Deep CNN-based detection and temporal sequence modelling	Accuracy, Precision, Recall, F1-score reported	Technical; Post-hoc	Experimental evaluation and baseline comparison
45	[55]	Integrated match analysis: detection, tracking, shot classification	Elite match videos	Broadcast footage; VideoBadminton dataset	Modified YOLOv8; court-aware tracking; SlowFast + TimeSformer	mAP@0.5 94.12%; Top-1 77.8%; Top-5 94.24%; 28 FPS	Technical and Tactical; Real-time and Post-hoc	Train/Test split; baseline comparison; confusion matrix

46	[56]	Real-time action recognition and pose estimation with shot refinement	Professional match videos (62 matches)	Broadcast footage with annotated hits	Improved YOLOv8 + 13-keypoint pose model + Kalman filtering	Accuracy 90.2%; F1-score 87.5%; Temporal error 2.14 frames	Technical and Tactical; Real-time and Post-hoc	Baseline comparison; cross-scenario evaluation
47	[57]	Stroke recognition using Quantum CNN (QCNN)	Competitive players (540 stroke samples)	Broadcast videos; OpenPose joint data	QCNN vs SVM and 3D-CNN	Accuracy and F1 up to 0.965; noise robustness ACC 0.882	Technical; Post-hoc	70/15/15 split; baseline comparison; noise evaluation
48	[58]	Deep learning–based badminton action recognition with multimodal fusion and computational optimisation	19 university-level athletes (VideoBadminton dataset; 7,822 clips)	VideoBadminton dataset (annotated match videos)	3D-CNN, Bi-LSTM, Video Swin Transformer; multimodal fusion; transfer learning	Accuracy up to 0.967; F1 up to 0.953; inference time 10.8–13.1 ms; reduced FLOPs and memory usage	Technical; Real-time and Post-hoc	70/15/15 split; comparison with temporal graph network and other baseline models; efficiency and performance benchmarking
49	[59]	Badminton action recognition with multimodal fusion and computational optimisation	19 university athletes (VideoBadminton dataset; 7,822 clips; 18 action classes)	Annotated match videos (VideoBadminton)	3D-CNN, Bi-LSTM, Video Swin Transformer; multimodal fusion; transfer learning	Accuracy up to 0.967; F1 up to 0.953; MSE 0.024–0.037; inference 10.8–13.1 ms	Technical; Real-time and Post-hoc	70/15/15 split; comparison with temporal graph convolutional network and other multimodal networks; efficiency benchmarking
50	[60]	Key frame detection in badminton swings for educational feedback	University students (102 videos; 143 in intervention)	Tablet video; MediaPipe 3D skeleton data	ST-GCN; MLP; temporal correction algorithm	Recall; F1-score; Average Positional Error	Technical; Real-time and Post-hoc	Cross-validation; intervention study; statistical evaluation
51	[3]	Tactical badminton analysis using computer vision and Retrieval-Augmented Generation (RAG)-anhanced large language model (LLM)	Elite singles matches (>750 rallies)	1080p broadcast videos (BWF)	YOLOv8; MediaPipe pose; multi-cue shot detection; RAG-LLM (LangChain + FAISS)	mAP 0.94 (player); 0.83 (shuttle); F1 0.92	Tactical; Post-hoc	Quantitative evaluation; statistical testing; expert validation

Table II shows the range of research designs in AI-based badminton player assessment. Most studies employ controlled experiments to analyse technical execution, biomechanics, and tactics, though often with reduced ecological validity. Observational approaches using broadcast or archived footage capture authentic match play but are limited by video quality. A smaller number of works focus on algorithmic innovation or feasibility studies addressing deployment challenges. The table also shows a concentration on elite and collegiate athletes, with fewer studies on recreational or youth groups, such as Wang et al. [9] for elite tactical analysis, Sinadia and Murwantara [6] for collegiate profiling, Liu and Liang [11] for recreational cohorts, and Amudhan et al. [41] for youth shuttlecock detection. This emphasis provides insights for high-performance contexts but limits generalisability to the wider badminton community, which is dominated by recreational and developing players [28,41,43].

Most AI-based badminton player assessment studies rely on video footage as the primary data source [3,5,7–9], reflecting its accessibility through broadcasts and archives and its minimal burden on players. A smaller set employs wearable sensors to capture biomechanical or physiological data [51], with only a few combining video and sensors to leverage spatial–temporal resolution alongside on-body kinematics. For instance, Wang et al. [9] introduced the ShuttleSet dataset with annotated video for tactical analysis, Van Herbruggen et al. [13] integrated inertial and ultra-wideband sensors with video to study match strategies, and Zheng and Chen [32] fused video, sensors, and historical records to support both tactical assessment and injury prevention. While video remains the most practical modality, hybrid approaches offer promising opportunities to enhance accuracy and contextual richness in future badminton analytics.

3.1.3. AI Methodologies for Badminton Player Assessment

AI research in badminton can be categorised into four principal application areas: stroke analysis, movement-pattern analysis, spatio-temporal or tactical analysis, and multi-modal technical evaluation. The literature is predominantly concentrated on stroke-level modelling, where computer vision, deep learning, and pose estimation techniques are applied to detect and classify strokes from video data. Movement-pattern studies focus on footwork recognition, trajectory tracking, and positional dynamics to characterise court coverage and recovery behaviour. Spatio-temporal or tactical approaches examine rally sequences, shot influence, and strategic decision-making processes, extending analysis beyond isolated actions to match-level contexts. Multi-modal technical evaluation integrates heterogeneous data sources, including video, wearable sensors, and biomechanical signals, to provide more comprehensive performance assessment. Overall, current research emphasises visual stroke detection, while integrated and strategy-oriented evaluation frameworks remain comparatively less developed.

Stroke analysis tasks in badminton frequently adopt deep learning classifiers such as 2D- CNNs, 3D-CNNs, and transformer-based sequence models to recognise stroke categories including smashes, drops, clears, and drives [5,18,30,39]. CNN-based approaches excel at capturing spatial features from individual frames, while 3D-CNNs extend this to spatio-temporal volumes, improving recognition of subtle motion cues. Transformer architectures and temporal CNNs further enhance performance by modelling long-range dependencies across rally sequences, making them effective in distinguishing strokes with similar visual appearance but different execution timing. Reported accuracies often exceed 85–90%, highlighting the maturity of these models for technical skill classification [10,13,17,18,23,29]. Movement-pattern analysis, on the other hand, builds on pose estimation outputs from frameworks such as OpenPose and MediaPipe, which generate joint keypoints used as input features for downstream models [10–12,31]. Recurrent neural networks, long short-term memory (LSTM), and graph convolutional networks are commonly employed to capture temporal dependencies and relational dynamics between body joints, enabling assessment of agility, footwork recovery, and tactical positioning [22]. Some studies incorporate You Only Look Once (YOLO)-based detectors for robust player localisation before pose extraction, ensuring consistent tracking under broadcast video conditions [9,12,18,28,31].

Movement-pattern and rally-sequencing analyses in badminton performance span spatial, temporal, and cognitive domains, including court coverage [22,48], footwork sequencing [12,23], recovery positioning [10], and contextual rally dynamics [7–9]. Spatial mapping and trajectory overlays produce heat-map visualisations aligned with expert evaluations [12,23], while pose-estimation models reliably classify six-corner footwork patterns [9]. Recovery-position models based on skeletal features achieve high accuracy but vary in predictive stability, reflecting the situational complexity of tactical decision-making [10]. Rally-sequencing studies highlight strategic shot-to-shot transitions that distinguish playing styles and outcomes [7–9], with machine learning approaches including decision tree (DT), random forest (RF), and graph models used to evaluate shot selection in rally contexts [7,22]. Integrating perceptual–cognitive markers, Tan and Teoh [27] combined Quiet Eye metrics with biomechanical features in a neural network, achieving over 85% accuracy in predicting shot outcomes and identifying gaze duration and onset timing as critical predictors.

3.1.4. Assessment Validation Approaches

AI-based badminton player assessment employs four main categories of performance indicators: biomechanical, shot quality, tactical, and movement-related metrics. Biomechanical measures, such as joint angles and segment velocities, are validated against expert annotations and biomechanical reference models, and these features are subsequently encoded as kinematic vectors for classifier training [10,12,40]. Shot-quality indicators, including accuracy, consistency, and power, are benchmarked using supervised classification pipelines with cross-validation, and they function as target variables in stroke recognition models [26]. Tactical metrics, such as shot selection and court positioning, are assessed through predictive accuracy and correlations with rally outcomes [55], and they are typically modelled as sequential dependencies within recurrent or transformer-based architectures [25]. Movement-related measures, notably footwork efficiency and court coverage, are validated using pose-tracking precision and trajectory error, and they are structured into graph or temporal representations for learning locomotor strategies [12].

Within AI-based badminton assessment, existing frameworks vary according to their analytical focus and temporal orientation. Most are designed as technical post-analysis systems, emphasising stroke mechanics and biomechanical precision once matches have concluded [7,17]. A smaller body of work addresses real-time technical assessment, where immediate feedback is achieved under favourable computational conditions [40]. Tactical strategy frameworks are reported less frequently and are generally retrospective, analysing rally sequences and decision-making behaviour through post-match data [8]. Real-time tactical guidance systems are rare, though studies demonstrate their capacity to provide adaptive decision support during active play [9,12,23,38]. Hybrid frameworks that combine technical and tactical elements across flexible timelines are limited but illustrate attempts to capture mechanical precision alongside contextual dynamics [24,28].

3.2. Common Challenges in AI-Based Badminton Assessment Research (RQ 2)

A recurrent theme across the reviewed literature is the persistence of technical challenges that constrain the reliability and scalability of AI-based badminton assessment systems. Data acquisition difficulties are the most frequently reported, particularly in relation to shuttlecock tracking and court reference extraction. Hsu et al. [40] highlighted the limitations of monocular video, where occlusion and high shuttle speed hinder consistent localisation, thereby impairing downstream analytical processes. Similarly, Wei and Weng [37] demonstrated that court line extraction from broadcast footage remains sensitive to lighting variation and camera zoom, reducing the stability of detection pipelines. These examples underscore how the quality and consistency of raw input data continue to represent a major obstacle for performance assessment frameworks.

Integration complexity emerges as another common limitation, particularly in systems that combine multiple data modalities. Ghosh et al. [28] emphasised the engineering burden of synchronising heterogeneous streams within the DeCoach framework, while Van Herbruggen et al. [13] noted similar challenges when aligning inertial sensor data with high-frame-rate video to maintain biomechanical fidelity. Closely related are concerns of generalisability, as many models perform well within narrowly defined datasets but degrade when applied to different playing styles or populations. Baclig et al. [34] demonstrated this difficulty when multi-player tracking models developed in squash contexts lost accuracy in badminton rallies, and Zhi et al. [39] reported analogous domain-shift effects in stroke recognition tasks.

Further limitations involve computational efficiency, real-time processing, and tracking precision. Yang et al. [18] explored model pruning and mixed-precision computation to preserve analytic accuracy while remaining within hardware constraints, whereas Salim et al. [26] designed adaptive inference pipelines that modulate processing load according to rally tempo. Tracking accuracy has also been identified as a recurring difficulty, as even minor errors in court-line detection can propagate across extended rallies, producing cumulative distortions in positional analysis [38,39]. These technical challenges illustrate the ongoing tension between methodological sophistication, practical feasibility, and ecological validity in AI-based badminton performance assessment.

3.3 Interpretability and Usability of AI Outputs in Badminton Player Assessment (RQ3)

3.3.1 Model Interpretability and Explainable Vision–Language Frameworks

Only 4 of the 51 reviewed studies incorporated explicit interpretability mechanisms. Wang [8] provided frame-level predictions with Grad-CAM heat maps that highlight the court regions influencing each decision, whereas Chen et al. [9] attached SHAP value attributions to rally-phase classifications. These visual or feature-based explanations transform otherwise opaque predictions into reasoned assessments that align with coaching terminology. Despite their demonstrated value, the limited uptake indicates a substantial gap between algorithmic precision and user confidence.

More recently, interpretability has also been extended beyond feature attribution towards structured vision–language reasoning. The ChatMatch framework [48] integrates meta-feature extraction, rule-based knowledge decoding, and LLM agents to generate structured and unstructured explanations of match events. By transforming spatial, action, and gesture recognition outputs into descriptive rally narratives and statistical summaries, the system provides traceable reasoning pathways between visual evidence and analytical conclusions [3]. This hybrid architecture illustrates a shift from post hoc visualisation towards interactive, explainable inference mechanisms that support professional query-based analysis.

3.3.2 Implications for Coaching Practice

Evidence from practice-focused evaluations shows that interpretable outputs improve both usability and acceptance. DeCoach [28] is used to quantify vertical growth stages objectively. Wang et al. [29] integrated bounding-box projections from their YOLO-HGNet detector with kinematic graphs, allowing for inspection of fast exchanges that are difficult to analyse visually. At larger scales, the ShuttleSet framework introduced by Wang et al. [22] processed more than thirty-six thousand strokes and provides dashboard summaries that shorten the routine assessment of approximately five hundred players. In a complementary line of work, Zhang and Zhong [45] developed a Kinect-based auxiliary training system that integrates multi-sensor skeletal fusion, gesture recognition, and hierarchical action evaluation to support structured badminton teaching. Their system demonstrated high motion-recognition accuracy and enabled real-time corrective feedback, illustrating how sensor-driven pose analysis can be embedded into pedagogical workflows.

3.3.3 Applications for Competition Organisation

Explainable assessment outputs also facilitate fairer competition structures. Sinadia and Murwantara [6] demonstrated that machine-learning profiles linked to positional heat maps can refine club assessment systems, while Asriani et al. [17] proved that spatio-temporal rally analysis is more sensitive than fixed-period reviews when reclassifying players. Xie et al. [44] further reported that pose-aware analytics enable organisers to design tournaments tailored to diverse tactical styles. Collectively, these studies suggest that transparent analytics support equitable seeding, responsive regrading, and data-informed scheduling.

3.3.4 Player Development Implications

For athletes, interpretable feedback promotes self-directed improvement. A study by Liu and Liang [11] presents an action-recognition tool that players use between formal sessions to monitor technique independently. Yuan et al. [7] proposed a multi-feature framework that blends technical and tactical indices to personalise training priorities. Tan and Teoh [27] combined neural network shot predictions with accurate metrics, providing objective evidence of skill progression that can enhance motivation. These findings indicate that accessible explanations can extend the impact of AI systems beyond coaching sessions, supporting continuous learning and goal-oriented practice.

3.3.5 Methodological Effectiveness

Many of the studies reviewed highlight recurring themes in performance testing that are increasingly relevant to practical coaching. One area with strong potential is the use of pose estimation to analyse player movements. For example, the enhanced YOLO model developed by Yang et al. [18] improved the accuracy of posture detection in badminton players by more than eight percent compared to earlier systems. This improvement was achieved by adjusting how the model processes visual information, resulting in greater efficiency and reliability. However, the availability of badminton-specific datasets remains limited, particularly for multi-view capture and synchronised multi-modal data [23], and addressing these constraints is essential for achieving more robust and generalisable pose-estimation models.

In addition to technical evaluation, some studies have explored the use of spatial and temporal features to support player assessment, including measurements of body position and timing. These models offer promising frameworks for translating movement data into meaningful performance insights. However, evaluating tactical decision-making remains more challenging. As shown in the ShuttleSet study by Wang [22], analysing strategic behaviour requires substantial human annotation and expert interpretation. This makes tactical assessment harder to scale and apply consistently. These limitations point to the need for stronger collaboration between AI developers and coaching practitioners to create systems that are not only accurate but also practical and intuitive for use in real sport settings.

4. Discussion

This review set out to clarify three questions: which AI techniques work best for badminton player assessment, what methodological hurdles still stand in the way, and how clearly current systems communicate their findings. Thirty-four primary studies, summarised in Table II, show that computer vision pipelines now dominate the field. Yet, they sit alongside movement-pattern analytics, spatio-temporal sequence models, and a small but important group of multi-modal systems. Together, these works sketch a discipline that has matured rapidly but remains uneven in scope and depth.

Modern vision models that track body landmarks and shuttle flight now achieve remarkable accuracy, with Yang et al. [18] reporting stroke-recognition rates above 95%, demonstrating rapid progress in automated analysis. By combining image cues with temporal contexts, these systems can assess not only what players do but also when and why actions matter within a rally, providing coaches with quantitative insights into technique, tactical rhythm, and fatigue. Yet, despite matching or exceeding human reliability in routine evaluations, many studies acknowledge that performance often degrades under varying recording conditions, highlighting the gap between technical promise and practical robustness and raising the critical question of what barriers limit broader adoption.

Data collection emerges as the most frequently reported obstacle in AI-based badminton player assessment. Hsu et al. [40] observed that the shuttlecock often disappears from a single camera's field of view due to its high velocity, disrupting otherwise reliable analytic pipelines. Integration complexity represents another recurrent challenge, with Ghosh et al. [28] documenting the substantial effort needed to synchronise video streams, inertial sensors, and match statistics within the DeCoach framework. Additional constraints arise from limited generalisability, computational overhead, and real-time latency, all of which highlight the contrast between controlled laboratory benchmarks and the variable conditions of competition environments characterised by fluctuating lighting, multiple camera angles, and heterogeneous player styles.

Only two studies, Wang [8] and Chen et al. [9], integrated explanation tools such as Grad-CAM or SHAP, showing that even simple visual cues boost coach confidence and player engagement. More recent systems, including ChatMatch [48] and Court-to-Conversation [3], integrated RAG and LLM components to transform detected visual events into structured, query-driven analytical feedback. By grounding language outputs in retrieved match data and recognised actions, these frameworks establish traceable links between raw visual evidence and narrative explanations, thereby supporting coach-player interaction. Future progress depends on open multi-modal datasets, models with domain adaptation and uncertainty estimation, and interpretability as a standard rather than a novelty. Longitudinal studies that track how AI-feedback shapes skill development and performance will then provide the evidence needed for widespread adoption.

5. Conclusion

This systematic review examined the current development of AI applications in badminton player assessment, highlighting both methodological diversity and several key limitations. Although AI systems offer the potential for objective, consistent, and multi-dimensional performance evaluation, their adoption remains limited due to weak theoretical grounding, insufficient methodological rigour, and limited contextual adaptability. Existing studies employ various approaches, including computer vision for stroke recognition, movement analysis, spatio-temporal modelling, and multi-modal frameworks. Future research should strengthen the theoretical foundation by linking computational outputs with established models of skill acquisition and athlete development, while also developing standardised performance metrics to enable meaningful comparisons across studies. A promising future trajectory lies in developing interactive and explainable AI systems supported by LLMs, enabling performance analysis to be translated into natural, context-aware explanations that coaches and players can easily interpret and utilise.

Declarations

Competing Interests

The authors declare that they have no known competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

Funding

The authors received no specific funding for this work.

Author Contribution

R.C. and F.U.D. designed the study and supervised a student group in carrying out the initial analysis. C.A. carried out further analysis and wrote the manuscript. All authors reviewed, edited and approved the final manuscript.

Acknowledgement

The authors would like to acknowledge the initial work carried out by a student group led by Utsab Gyawali, which contributed to the early exploration of this research topic.

References

1. Wörner EA, Safran MR. Racquet Sports: Tennis, Badminton, Racquetball, Squash. Specific Sports-Related Injuries, Cham: Springer International Publishing; 2021, p. 431–46. https://doi.org/10.1007/978-3-030-66321-6_30.
2. Woo T-MT, Alam F. Comparative aerodynamics of synthetic badminton shuttlecocks. Sports Engineering 2018;21:21–9. https://doi.org/10.1007/s12283-017-0241-2.
3. Bharadwaj K, Srinivasa G. Court to conversation: Tactical badminton analysis via computer vision and RAG-enhanced LLMs. Knowl Based Syst 2026;333:115027. https://doi.org/10.1016/j.knosys.2025.115027.
4. Jekauc D, Burkart D, Fritsch J, Hesenius M, Meyer O, Sarfraz S, et al. Recognizing affective states from the expressive behavior of tennis players using convolutional neural networks. Knowl Based Syst 2024;295:111856. https://doi.org/10.1016/j.knosys.2024.111856.
5. Ghosh A, Singh S, Jawahar C V. Towards structured analysis of broadcast badminton videos. Proceedings of the IEEE Winter Conference on Applications of Computer Vision 2018:296–304. https://doi.org/10.1109/WACV.2018.00039.
6. Sinadia HE, Murwantara IM. Sports analytics: A comparison of machine learning performance for profiling badminton athlete. Proceedings of the International Conference on Technology Innovation and Its Applications 2022. https://doi.org/10.1109/ICTIIA54654.2022.9935852.
7. Yuan H, Wang Y, Yang K, Bin Y. Prediction model and technical and tactical decision analysis of women's badminton singles based on machine learning. PLoS One 2024;19. https://doi.org/10.1371/journal.pone.0312801.
8. Wang KD. Enhancing badminton player performance via a closed-loop AI approach: Imitation, simulation, optimization, and execution. Proceedings of the International Conference on Information and Knowledge Management 2023:5189–92. https://doi.org/10.1145/3583780.3616001.
9. Chen WT, Wu HY, Shih YA, Wang CC, Wang YS. Exploration of player behaviours from broadcast badminton videos. Computer Graphics Forum 2023;42:267–79. https://doi.org/10.1111/cgf.14786.
10. Krishnaram N, Ahamed J, Sathyamoorthy N, Sandaruwan KD, Athapaththu AMKB. Skeletal point analysis to determine the accuracy of forehand smash shots played by badminton players. J Natl Sci Found 2024;52:125–42. https://doi.org/10.4038/jnsfsr.v52i1.12141.
11. Liu J, Liang B. An action recognition technology for badminton players using deep learning. Mobile Information Systems 2022. https://doi.org/10.1155/2022/3413584.
12. Jannet K, Sunar MS, Molla MMI, Bin As'Ari MA. A deep learning approach to badminton player footwork detection based on YOLO models: A comparative study. Proceedings of the IEEE International Conference on Signal and Image Processing Applications 2024. https://doi.org/10.1109/ICSIPA62061.2024.10686537.
13. Van Herbruggen B, Fontaine J, Simoen J, De Mey L, Peralta D, Shahid A, et al. Strategy analysis of badminton players using deep learning from IMU and UWB wearables. Internet of Things 2024;27. https://doi.org/10.1016/j.iot.2024.101260.
14. Marques A, Gouveira ÉR, Peralta M, Martins J, Venturini J, Henriques-Neto D, et al. Cardiorespiratory fitness and telomere length: a systematic review. J Sports Sci 2020;38:1690–7. https://doi.org/10.1080/02640414.2020.1754739.
15. Zhou D, Keogh JWL, Ma Y, Tong RKY, Khan AR, Jennings NR. Artificial intelligence in sport: A narrative review of applications, challenges and future trends. J Sports Sci 2025:1–16. https://doi.org/10.1080/02640414.2025.2518694.
16. Vickery-Howe DM, Bonanno DR, Dascombe BJ, Drain JR, Clarke AC, Hoolihan B, et al. Physiological, perceptual, and biomechanical differences between treadmill and overground walking in healthy adults: A systematic review and meta-analysis. J Sports Sci 2023;41:2088–120. https://doi.org/10.1080/02640414.2024.2312481.
17. Asriani F, Azhari A, Wahyono W. Improving badminton action recognition using spatio-temporal analysis and a weighted ensemble learning model. Computers, Materials and Continua 2024;81:3079–96. https://doi.org/10.32604/cmc.2024.058193.
18. Yang W, Jiang M, Fang X, Shi X, Guo Y, Al-qaness MAA. A high-precision and efficient method for badminton action detection in sports using You Only Look Once with Hourglass Network. Eng Appl Artif Intell 2024;137:109177. https://doi.org/10.1016/j.engappai.2024.109177.
19. Cao Z, Liao T, Song W, Chen Z, Li C. Detecting the shuttlecock for a badminton robot: A YOLO based approach. Expert Syst Appl 2021;164:113833. https://doi.org/10.1016/j.eswa.2020.113833.
20. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021:n71. https://doi.org/10.1136/bmj.n71.
21. Helbach J, Hoffmann F, Pieper D, Allers K. Reporting according to the preferred reporting items for systematic reviews and meta-analyses for abstracts (PRISMA-A) depends on abstract length. J Clin Epidemiol 2023;154:167–77. https://doi.org/10.1016/j.jclinepi.2022.12.019.
22. Wang WY, Huang YC, Ik TU, Peng WC. ShuttleSet: A human-annotated stroke-level singles dataset for badminton tactical analysis. Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2023:5126–36. https://doi.org/10.1145/3580305.3599906.
23. Luo J, Hu Y, Davids K, Zhang D, Gouin C, Li X, et al. Vision-based movement recognition reveals badminton player footwork using deep learning and binocular positioning. Heliyon 2022;8. https://doi.org/10.1016/j.heliyon.2022.e10089.
24. Ding N, Takeda K, Fujii K. Deep reinforcement learning in a racket sport for player evaluation with technical and tactical contexts. IEEE Access 2022;10:54764–72. https://doi.org/10.1109/ACCESS.2022.3175314.
25. Ashfaq F, Jhanjhi NZ, Khan NA. Badminton player's shot prediction using deep learning. Lecture Notes in Bioengineering 2023:233–43. https://doi.org/10.1007/978-981-99-0297-2_19.
26. Salim MA, Williamdy W, Steven E. AI-powered badminton shot classification. Proceedings of the International Conference on Technology Innovation and Its Applications 2024. https://doi.org/10.1109/ICTIIA61827.2024.10761547.

27. Tan S, Teoh TT. Predicting shot accuracy in badminton using quiet eye metrics and neural networks. Applied Sciences 2024;14:9906. https://doi.org/10.3390/app14219906.
28. Ghosh I, Ramasamy Ramamurthy S, Chakma A, Roy N. DeCoach: Deep learning-based coaching for badminton player assessment. Pervasive Mob Comput 2022;83. https://doi.org/10.1016/j.pmcj.2022.101608.
29. Wang WY, Chan TF, Peng WC, Yang HK, Wang CC, Fan YC. How is the stroke? Inferring shot influence in badminton matches via long short-term dependencies. ACM Trans Intell Syst Technol 2022;14. https://doi.org/10.1145/3551391.
30. Peng W, Zheng W. An AI-based badminton smash detection using residual-shuffle network optimized based on upgraded pufferfish optimizer. Ain Shams Engineering Journal 2025;16:103414. https://doi.org/10.1016/j.asej.2025.103414.
31. Wu Y, Chen Z, Zhang H, Yang Y, Yi W. Enhanced Pose Estimation for Badminton Players via Improved YOLOv8-Pose with Efficient Local Attention. Sensors 2025;25:4446. https://doi.org/10.3390/s25144446.
32. Zheng Y, Chen E. Harnessing deep learning and advanced analytics to revolutionise badminton performance. International Journal of Information and Communication Technology 2025;26. https://doi.org/10.1504/IJICT.2025.10070826.
33. Rahmad NA, Sufri NAJ, Muzamil NH, As'ari MA. Badminton player detection using faster region convolutional neural network. Indonesian Journal of Electrical Engineering and Computer Science 2019;14:1330–5. https://doi.org/10.11591/ijeecs.v14.i3.pp1330-1335.
34. Baclig MM, Ergezinger N, Mei Q, Gul M, Adeeb S, Westover L. A deep learning and computer vision based multi-player tracker for squash. Applied Sciences 2020;10. https://doi.org/10.3390/app10248793.
35. Liu P, Wang J-H. MonoTrack: Shuttle trajectory reconstruction from monocular badminton video. Proceedings of the IEEE Computer Vision and Pattern Recognition Workshops 2022:3512–21. https://doi.org/10.1109/CVPRW56347.2022.00395.
36. Wu F, Wang Q, Bian J, Ding N, Lu F, Cheng J, et al. A survey on video action recognition in sports: Datasets, methods and applications. IEEE Trans Multimedia 2023;25:7943–66. https://doi.org/10.1109/TMM.2022.3232034.
37. Wei CT, Weng SK. A court line extraction algorithm for badminton tournament videos with horizontal line projection learning. IET Image Process 2023;17:2907–24. https://doi.org/10.1049/ipr2.12838.
38. Zhang L, Dai H. Motion trajectory tracking of athletes with improved depth information-based KCF tracking method. Multimed Tools Appl 2023;82:26481–93. https://doi.org/10.1007/s11042-023-14929-6.
39. Zhi J, Sun Z, Zhang R, Zhao Z. Badminton video action recognition based on time network. Journal of Computational Methods in Sciences and Engineering 2023;23:2739–52. https://doi.org/10.3233/JCM-226889.
40. Hsu YH, Yu CC, Cheng HY. Enhancing badminton game analysis: An approach to shot refinement via a fusion of shuttlecock tracking and hit detection from monocular camera. Sensors 2024;24. https://doi.org/10.3390/s24134372.
41. Amudhan AN, Vrajesh SR, Sudheer AP, Lijiya A. RFSOD: A lightweight single-stage detector for real-time embedded applications to detect small-size objects. J Real Time Image Process 2022;19:133–46. https://doi.org/10.1007/s11554-021-01170-3.
42. Tan YE, See J, Abdullah J, Wang LK, Ban KW. A deep learning based framework for badminton rally outcome prediction. Proceedings of the International Symposium on Intelligent Signal Processing and Communication Systems 2022. https://doi.org/10.1109/ISPACS57703.2022.10082764.
43. Gonzalez MR, Sandoval JC, Tostado GF, Santamaria SS, San Blas HS, Gonzalez G V, et al. Development of a stroke detection system for racket sports. Proceedings of the International Conference on Disruptive Technologies, Tech Ethics, and Artificial Intelligence 2023:320–8. https://doi.org/10.1007/978-3-031-38344-1_31.
44. Xie Y, Xie X, Xia H, Zhao Z. Algorithmic study on position and movement method of badminton doubles. Sci Program 2022. https://doi.org/10.1155/2022/6422442.
45. Zhang L, Zhong P. Feather teaching and training based on Kinect sensor and gesture recognition technology. Systems and Soft Computing 2025;7:200349. https://doi.org/10.1016/j.sasc.2025.200349.
46. Peng W, Zheng W. An AI-based badminton smash detection using residual-shuffle network optimized based on upgraded pufferfish optimizer. Ain Shams Engineering Journal 2025;16:103414. https://doi.org/10.1016/j.asej.2025.103414.
47. Jiang K. Optimization study of badminton sports training system based on MoileNet OpenPose lightweight human posture estimation model. Entertain Comput 2025;54:100975. https://doi.org/10.1016/j.entcom.2025.100975.
48. Zhang J, Han D, Han S, Li H, Lam W-K, Zhang M. ChatMatch: Exploring the potential of hybrid vision–language deep learning approach for the intelligent analysis and inference of racket sports. Comput Speech Lang 2025;89:101694. https://doi.org/10.1016/j.csl.2024.101694.
49. Li J, Fan Y, Chen R, Liang S, Feng Y, He Y, et al. An Intelligent Badminton Handle With Multinode MEMS Sensors for Explainable Motion Recognition. IEEE Internet Things J 2025;12:37022–34. https://doi.org/10.1109/JIOT.2025.3580545.
50. Shih C-Y, Lin Y-T, Chen W, Huang J-C. SVM-Adaboost based badminton offensive movement parsing technique. Signal Image Video Process 2025;19:280. https://doi.org/10.1007/s11760-025-03865-7.
51. Liu D, Che L, Qi F, Zhang K, Cao D. Evaluating the sports performance of badminton players based on grip strength of the real hitting scenario. Sci Rep 2026;16:5055. https://doi.org/10.1038/s41598-026-35596-1.
52. Chen E, Zheng Y. Harnessing deep learning and advanced analytics to revolutionise badminton performance. International Journal of Information and Communication Technology 2025;26:61–79. https://doi.org/10.1504/IJICT.2025.146097.
53. Liu M. Badminton Technical Training Analysis System Based on Intelligent Motion Tracking Decomposition Model. 2025 International Conference on Algorithm, Artificial Intelligence and Computer Vision (AAICV), IEEE; 2025, p. 01–4. https://doi.org/10.1109/AAICV66571.2025.00013.

54. Tan and J. Komar, “How Data Can Help You Beat Your Next Opponent: Applications of Artificial Intelligence for Performance Analysis in Badminton,” in Sports Data Analytics: Techniques, Applications, and Innovations, M. S. Raval, T. Kaya, N. S. Artan, and C. Taber, Eds. Singapore: Springer Nature Singapore, 2026, pp. 141–162. https://doi.org/10.1007/978-981-95-5132-3_8.
55. Singha KAK, Ali MA, Acharjee T, Debnath P, Ahmadullah S. Badminton Match Analysis Along With Players and Shuttlecock Setection Using Hybrid CNN. 2025 IEEE Silchar Subsection Conference (SILCON), IEEE; 2025, p. 1–6. https://doi.org/10.1109/SILCON67893.2025.11327275.
56. Luo L, Zhang L, Liu H. An integrated deep learning framework for real-time badminton action recognition and pose estimation with shot-level semantic refinement. Australian Journal of Electrical and Electronics Engineering 2025:1–20. https://doi.org/10.1080/1448837X.2025.2553948.
57. Zhu X, Liu L, Huang J, Chen G, Ling X, Chen Y. The analysis of motion recognition model for badminton player movements using machine learning. Sci Rep 2025;15:19030. https://doi.org/10.1038/s41598-025-02771-9.
58. Ma H, Zhang F, Liang N. Development and Training Strategy of Badminton Action Recognition System Under the Background of Artificial Intelligence. IEEE Access 2025;13:71577–85. https://doi.org/10.1109/ACCESS.2025.3563302.
59. Chang Y. A method of badminton video motion recognition based on adaptive enhanced AdaBoost algorithm. Int J Biom 2025;17:86–101. https://doi.org/10.1504/IJBM.2025.143715.
60. Hsu J-H, Lee C-C, Chang J-Y, Lee D-S. Key Frame Detection in Badminton Swings and Its Application to Physical Education. IEEE Access 2025;13:91248–62. https://doi.org/10.1109/ACCESS.2025.3572105.

Figures

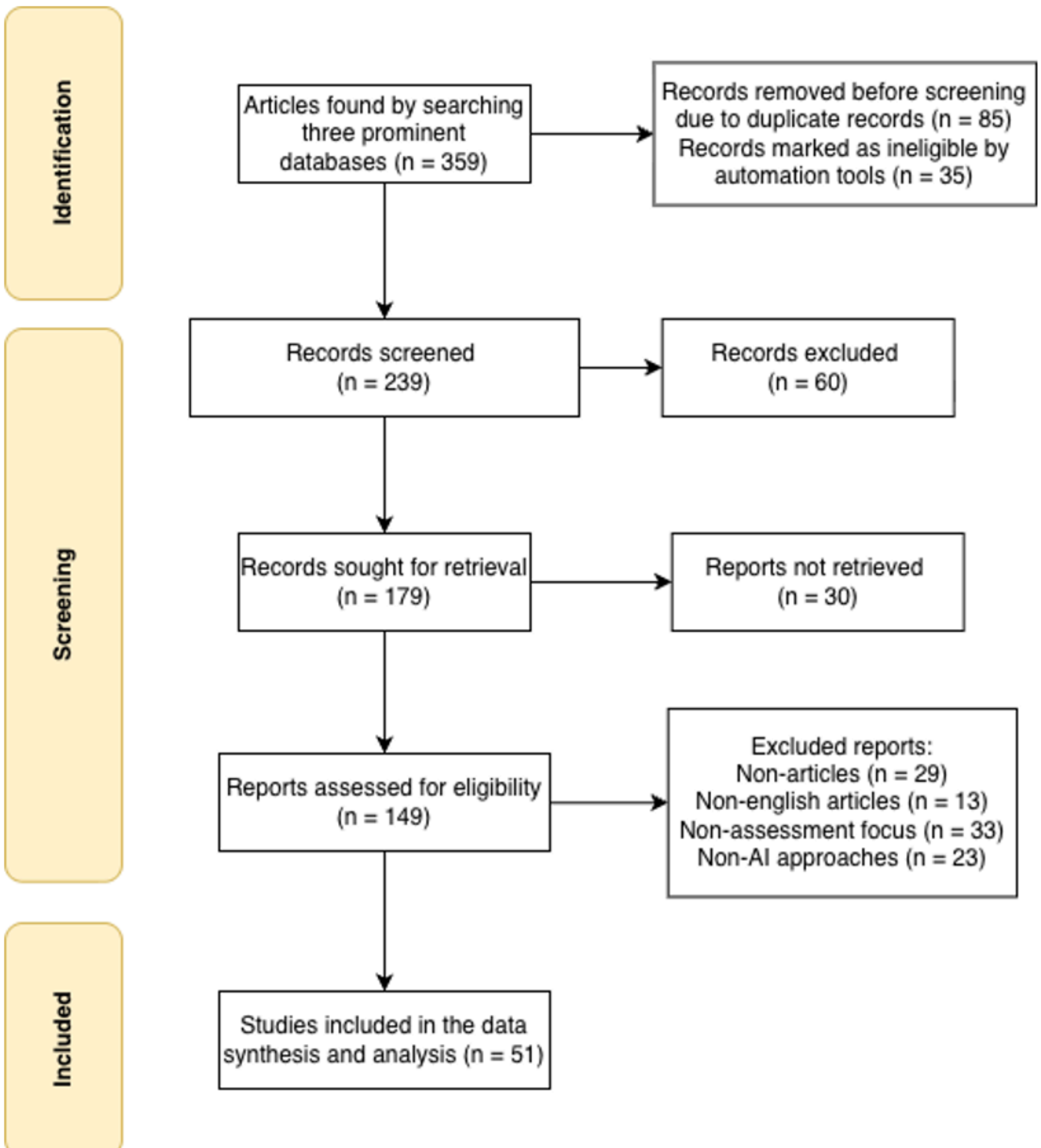

Figure 1

PRISMA flow diagram illustrating the systematic study selection process.

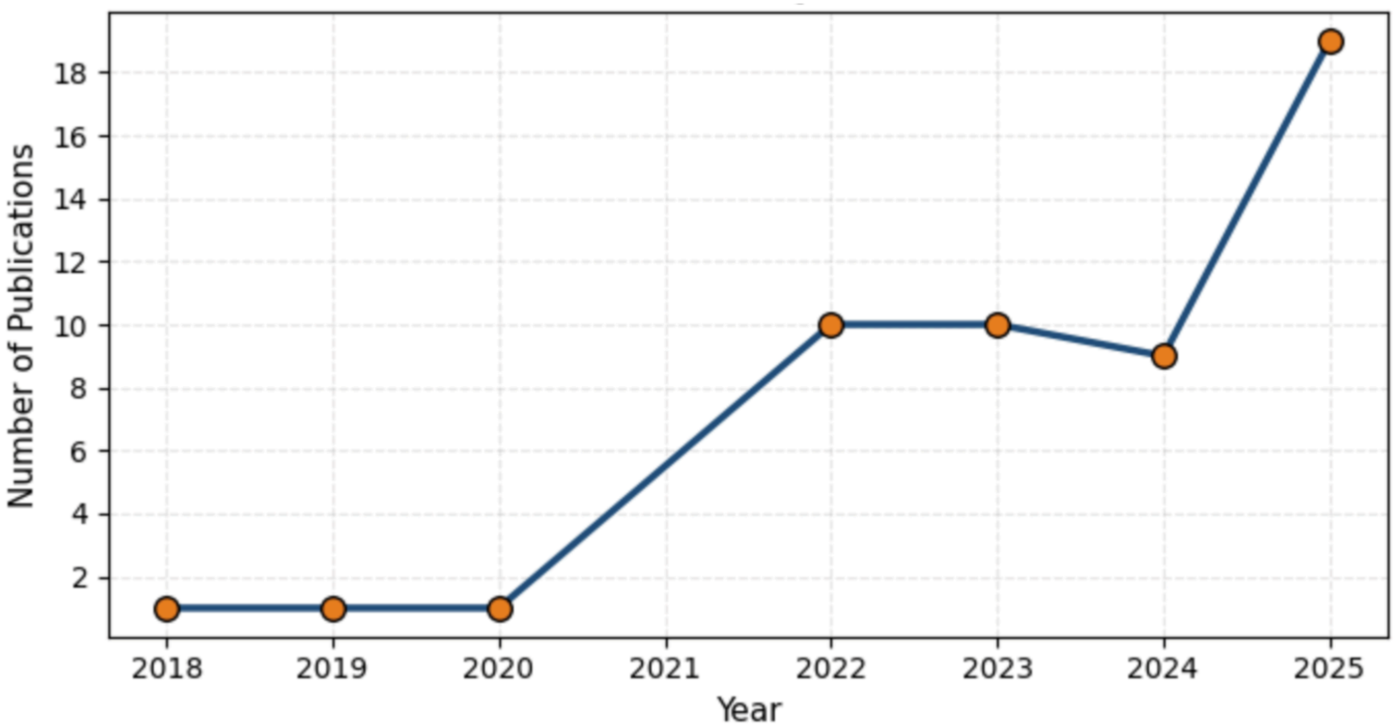

Figure 2

Annual publication trend of AI applications in badminton analytics from 2015 to 2025.

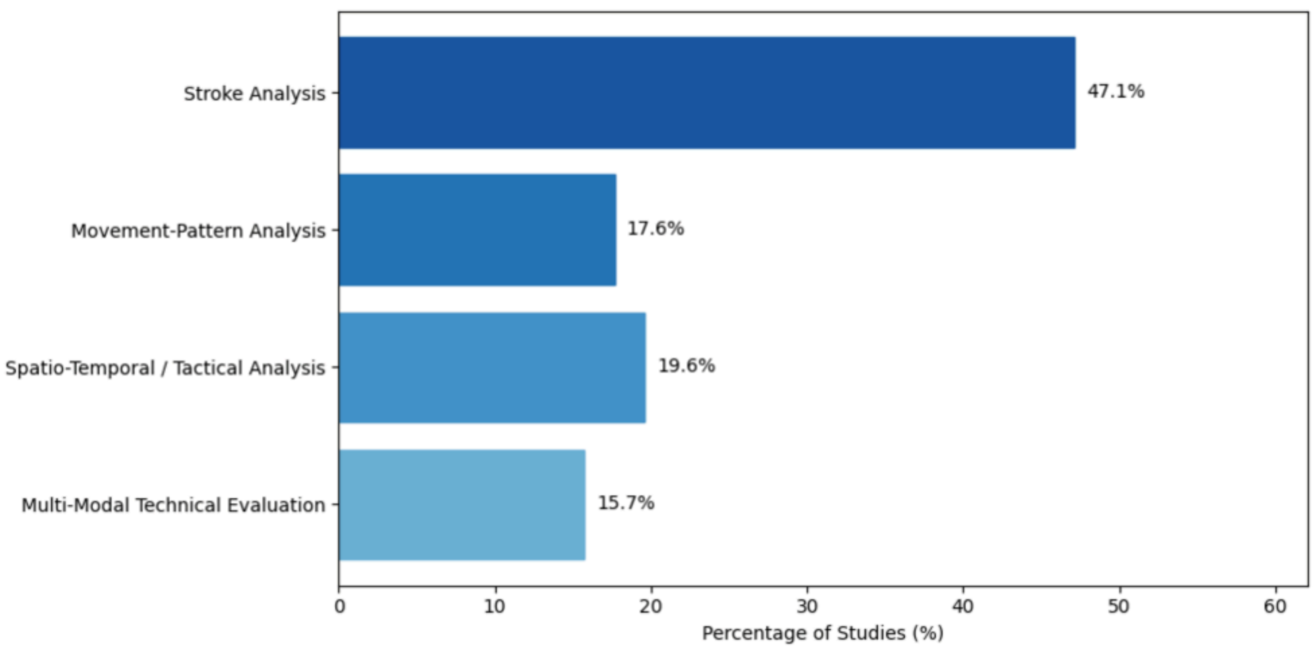

Figure 3

Distribution of methodological categories in AI-based badminton assessment studies.